

The $\text{SO}(3)$ – $\text{SU}(2)$ Correspondence and Quaternions

A complete derivation for the mathematical physicist,
with the answers a GNC or simulation engineer actually needs

Every algebraic step is written out. Nothing is left as “it can be shown.”

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1 How to read this, and the conventions we fix

This document has two interleaved tracks.

- The **main text** is the mathematics, done completely: definitions, propositions, and proofs in which every index manipulation and every series regrouping is displayed.
- The **blue boxes** translate the result just proved into the language of an engineer who must write, review, or debug flight code, and the **red boxes** record the specific way each result is gotten wrong in practice.

A mathematical physicist asked to support a GNC team is usually not asked “is the exponential map surjective?” They are asked “why did my quaternion flip sign in the middle of a maneuver?” The second question is the first question wearing work clothes. The boxes make that identification explicit.

1.1 Conventions, fixed once

Almost every contradiction between two quaternion references is a convention clash, not an error. We therefore fix everything now and never deviate.

- (C1) **Quaternion algebra: Hamilton.** Writing the quaternion units as $\mathbf{i}, \mathbf{j}, \mathbf{k}$ (reserving i for $\sqrt{-1}$): $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1$, hence $\mathbf{ij} = \mathbf{k}$, $\mathbf{jk} = \mathbf{i}$, $\mathbf{ki} = \mathbf{j}$.
- (C2) **Scalar first.** $q = (q_0, \mathbf{q}) = q_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k}$ with $\mathbf{q} = (q_1, q_2, q_3)^\top \in \mathbb{R}^3$.
- (C3) **Active rotations.** $R \in \text{SO}(3)$ rotates a vector within a fixed frame: $\mathbf{v}' = R\mathbf{v}$. It does *not* re-express a fixed vector in a rotated frame.
- (C4) **Sandwich acts on the left.** $\mathbf{v}' = q \mathbf{v} q^*$, with $\mathbf{v}$ embedded as the pure quaternion $(0, \mathbf{v})$.
- (C5) **Body-to-inertial.** When a physical attitude is meant, R and q carry body-frame coordinates to inertial coordinates: $\mathbf{v}_{\mathcal{I}} = R \mathbf{v}_{\mathcal{B}}$.
- (C6) **Pauli matrices.** $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Engineering translation: the convention table you should paste into your repository

The four independent binary choices, and what the two dominant camps pick:

Choice	Hamilton (this doc, Eigen, ROS)	JPL/Shuster (NASA attitude)
Component order	scalar first $(q_0, \mathbf{q})$	scalar last $(\mathbf{q}, q_4)$
Unit product	$\mathbf{ij} = +\mathbf{k}$	$\mathbf{ij} = -\mathbf{k}$
Composition	$R(p \otimes q) = R(p)R(q)$	$R(p \otimes q) = R(q)R(p)$
Default sense	body $\rightarrow$ inertial (active)	inertial $\rightarrow$ body (passive)

A “quaternion bug” is, in the overwhelming majority of cases, two modules disagreeing on one row of this table. Before debugging arithmetic, print $R(q)$ for $q = \exp$ of a 90° rotation about $\mathbf{e}_3$ and check the sign of the $(1, 2)$ entry. That single number distinguishes the conventions faster than any amount of code reading.

2 The three groups and their Lie algebras

2.1 $\mathrm{SO}(3)$ and $\mathfrak{so}(3)$

Definition 2.1. $\mathrm{SO}(3) = \{R \in \mathrm{GL}(3, \mathbb{R}) : R^\top R = I, \det R = +1\}$.

Proposition 2.2. $\mathfrak{so}(3) = \{A \in \mathbb{R}^{3 \times 3} : A^\top = -A\}$, a 3-dimensional real vector space.

Proof. Let $R(t)$ be a smooth curve in $\mathrm{SO}(3)$ with $R(0) = I$, and set $A = \dot{R}(0)$. Differentiating the constraint $R(t)^\top R(t) = I$ gives

$$\frac{d}{dt}(R^\top R) = \dot{R}^\top R + R^\top \dot{R} = 0.$$

Evaluating at $t = 0$, where $R(0) = I$,

$$A^\top + A = 0,$$

so A is skew-symmetric. Conversely, if $A^\top = -A$ then $R(t) = e^{tA}$ satisfies

$$R(t)^\top R(t) = e^{tA^\top} e^{tA} = e^{-tA} e^{tA} = I,$$

where the middle equality uses $A^\top = -A$ and the last uses that $-tA$ and tA commute. Also $\det e^{tA} = e^{t \operatorname{tr} A} = e^0 = 1$ because a skew matrix has zero diagonal, hence zero trace. So e^{tA} is a curve in $\mathrm{SO}(3)$ through I with velocity A . A skew 3×3 matrix is determined by its three entries above the diagonal, so $\dim \mathfrak{so}(3) = 3$. $\square$

Definition 2.3 (Hat map). For $\mathbf{v} = (v_1, v_2, v_3)^\top \in \mathbb{R}^3$ define

$$\widehat{\mathbf{v}} = \begin{pmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{pmatrix} \in \mathfrak{so}(3), \quad \text{equivalently} \quad (\widehat{\mathbf{v}})_{jk} = -\varepsilon_{jkl} v_l.$$

Lemma 2.4. $\widehat{\mathbf{v}} \mathbf{w} = \mathbf{v} \times \mathbf{w}$ for all $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$.

Proof. Componentwise, using $(\widehat{\mathbf{v}})_{jk} = -\varepsilon_{jkl} v_l$ and summing repeated indices,

$$(\widehat{\mathbf{v}} \mathbf{w})_j = (\widehat{\mathbf{v}})_{jk} w_k = -\varepsilon_{jkl} v_l w_k = \varepsilon_{jlk} v_l w_k = (\mathbf{v} \times \mathbf{w})_j,$$

where the third equality is the antisymmetry $\varepsilon_{jkl} = -\varepsilon_{jlk}$ and the fourth is the definition $(\mathbf{a} \times \mathbf{b})_j = \varepsilon_{jlk} a_l b_k$. Written out for $j = 1$: $(\widehat{\mathbf{v}} \mathbf{w})_1 = -v_3 w_2 + v_2 w_3 = v_2 w_3 - v_3 w_2$, which is indeed the first component of $\mathbf{v} \times \mathbf{w}$. $\square$

Take as basis $L_j = \widehat{\mathbf{e}}_j$, explicitly

$$L_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad L_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Proposition 2.5. $[L_1, L_2] = L_3$, and cyclically $[L_2, L_3] = L_1$, $[L_3, L_1] = L_2$. Equivalently $[L_j, L_k] = \varepsilon_{jkl} L_l$.

Proof. We compute L_1L_2 and L_2L_1 entry by entry rather than asserting the result. Writing rows, L_1 has rows $(0,0,0)$, $(0,0,-1)$, $(0,1,0)$ and L_2 has rows $(0,0,1)$, $(0,0,0)$, $(-1,0,0)$.

Row j of L_1L_2 is the combination of the rows of L_2 weighted by row j of L_1 :

$$\begin{aligned}\text{row 1} &: 0 \cdot (0,0,1) + 0 \cdot (0,0,0) + 0 \cdot (-1,0,0) = (0,0,0), \\ \text{row 2} &: 0 \cdot (0,0,1) + 0 \cdot (0,0,0) + (-1) \cdot (-1,0,0) = (1,0,0), \\ \text{row 3} &: 0 \cdot (0,0,1) + 1 \cdot (0,0,0) + 0 \cdot (-1,0,0) = (0,0,0),\end{aligned}$$

so $L_1L_2 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. Similarly row j of L_2L_1 is the rows of L_1 weighted by row j of L_2 :

$$\begin{aligned}\text{row 1} &: 0 \cdot (0,0,0) + 0 \cdot (0,0,-1) + 1 \cdot (0,1,0) = (0,1,0), \\ \text{row 2} &: (0,0,0), \\ \text{row 3} &: (-1) \cdot (0,0,0) + 0 + 0 = (0,0,0),\end{aligned}$$

so $L_2L_1 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. Subtracting,

$$[L_1, L_2] = L_1L_2 - L_2L_1 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = L_3.$$

The other two identities follow by cycling $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$, which permutes the L_j in the same way because $(\widehat{\mathbf{v}})_{jk} = -\varepsilon_{jkl}v_l$ is cyclically symmetric. $\square$

Corollary 2.6. $[\widehat{\mathbf{v}}, \widehat{\mathbf{w}}] = \widehat{\mathbf{v} \times \mathbf{w}}$.

Proof. Both sides are bilinear in $(\mathbf{v}, \mathbf{w})$, so it suffices to check basis vectors. With $\mathbf{v} = \mathbf{e}_j$, $\mathbf{w} = \mathbf{e}_k$ the left side is $[L_j, L_k] = \varepsilon_{jkl}L_l$ by Proposition 2.5, and the right side is $\widehat{\mathbf{e}_j \times \mathbf{e}_k} = \widehat{\varepsilon_{jkl}\mathbf{e}_l} = \varepsilon_{jkl}L_l$. $\square$

Engineering translation: angular velocity is simultaneously a vector and a matrix

Corollary 2.6 is the precise reason your code can carry $\boldsymbol{\omega}$ as a three-component array and still write $\dot{R} = R\widehat{\boldsymbol{\omega}}$. The hat map is a Lie algebra isomorphism $(\mathbb{R}^3, \times) \rightarrow (\mathfrak{so}(3), [\cdot, \cdot])$: cross product is matrix commutator in disguise. When a reviewer asks why you are allowed to move between the two, this is the theorem to cite, and it also tells you the failure mode — the identification is basis dependent, so it holds only in a right-handed orthonormal frame. In a left-handed frame every cross product flips sign.

2.2 $\text{SU}(2)$ and $\mathfrak{su}(2)$

Definition 2.7. $\text{SU}(2) = \{U \in \mathbb{C}^{2 \times 2} : U^\dagger U = I, \det U = 1\}$.

Proposition 2.8. $U \in \text{SU}(2)$ if and only if $U = \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix}$ with $\alpha, \beta \in \mathbb{C}$ and $|\alpha|^2 + |\beta|^2 = 1$. Consequently $\text{SU}(2)$ is diffeomorphic to the unit sphere $\mathbb{S}^3 \subset \mathbb{R}^4$.

Proof. Write $U = \begin{pmatrix} \alpha & \gamma \\ \beta & \delta \end{pmatrix}$. The condition $\det U = 1$ reads $\alpha\delta - \gamma\beta = 1$. For a 2×2 matrix of determinant 1 the inverse is $U^{-1} = \begin{pmatrix} \delta & -\gamma \\ -\beta & \alpha \end{pmatrix}$. Unitarity says $U^\dagger = U^{-1}$, and $U^\dagger = \begin{pmatrix} \bar{\alpha} & \bar{\beta} \\ \bar{\gamma} & \bar{\delta} \end{pmatrix}$. Equating entries:

$$\bar{\alpha} = \delta, \quad \bar{\beta} = -\gamma, \quad \bar{\gamma} = -\beta, \quad \bar{\delta} = \alpha.$$

The first and fourth are the same statement; so are the second and third. Substituting $\delta = \bar{\alpha}$ and $\gamma = -\bar{\beta}$ into $\det U = 1$ gives $\alpha\bar{\alpha} + \bar{\beta}\beta = |\alpha|^2 + |\beta|^2 = 1$. Writing $\alpha = a_0 + ia_1$, $\beta = a_2 + ia_3$ identifies the solution set with $\{a \in \mathbb{R}^4 : a_0^2 + a_1^2 + a_2^2 + a_3^2 = 1\} = \mathbb{S}^3$. $\square$

Proposition 2.9. $\mathfrak{su}(2) = \{X \in \mathbb{C}^{2 \times 2} : X^\dagger = -X, \operatorname{tr} X = 0\}$, a 3-dimensional real vector space with basis

$$E_j := -\frac{i}{2}\sigma_j, \quad j = 1, 2, 3.$$

Proof. Let $U(t) \subset \mathrm{SU}(2)$ with $U(0) = I$ and $X = \dot{U}(0)$. Differentiating $U^\dagger U = I$ at $t = 0$ gives $X^\dagger + X = 0$. Differentiating $\det U(t) = 1$ and using Jacobi's formula $\frac{d}{dt} \det U = \det U \cdot \operatorname{tr}(U^{-1} \dot{U})$ at $t = 0$ gives $\operatorname{tr} X = 0$. Conversely, if $X^\dagger = -X$ and $\operatorname{tr} X = 0$ then e^{tX} is unitary and has $\det e^{tX} = e^{t \operatorname{tr} X} = 1$.

For the basis: each σ_j is Hermitian and traceless, so $E_j = -\frac{i}{2}\sigma_j$ satisfies $E_j^\dagger = +\frac{i}{2}\sigma_j^\dagger = \frac{i}{2}\sigma_j = -E_j$ and $\operatorname{tr} E_j = -\frac{i}{2} \operatorname{tr} \sigma_j = 0$. A general anti-Hermitian traceless 2×2 matrix is $\begin{pmatrix} ic & z \\ -\bar{z} & -ic \end{pmatrix}$ with $c \in \mathbb{R}$, $z \in \mathbb{C}$, which is three real parameters, so the E_j (three real-independent elements) span. $\square$

Lemma 2.10 (Pauli product identity). $\sigma_j \sigma_k = \delta_{jk} I + i \varepsilon_{jkl} \sigma_l$, and consequently for any $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$,

$$(\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma}) = (\mathbf{a} \cdot \mathbf{b}) I + i (\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma}.$$

Proof. Direct computation on the six independent cases. Squares first:

$$\sigma_1^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I,$$

and identically $\sigma_2^2 = \sigma_3^2 = I$, matching $\delta_{jj} I$ with no ε term. Now the mixed products:

$$\sigma_1 \sigma_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i\sigma_3,$$

$$\sigma_2 \sigma_1 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} = -i\sigma_3.$$

So $\sigma_1 \sigma_2 = i\varepsilon_{12l} \sigma_l = i\sigma_3$ and $\sigma_2 \sigma_1 = i\varepsilon_{21l} \sigma_l = -i\sigma_3$, as claimed. Likewise

$$\sigma_2 \sigma_3 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} = i\sigma_1, \quad \sigma_3 \sigma_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\sigma_2,$$

and the reversed products give the opposite sign by the same arithmetic. This exhausts all j, k and establishes $\sigma_j \sigma_k = \delta_{jk} I + i \varepsilon_{jkl} \sigma_l$.

For the vector form, expand bilinearly:

$$(\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma}) = a_j b_k \sigma_j \sigma_k = a_j b_k (\delta_{jk} I + i \varepsilon_{jkl} \sigma_l) = (a_k b_k) I + i \varepsilon_{jkl} a_j b_k \sigma_l,$$

and $\varepsilon_{jkl} a_j b_k = \varepsilon_{ljk} a_j b_k = (\mathbf{a} \times \mathbf{b})_l$ by cyclicity of ε . $\square$

Proposition 2.11. $[E_j, E_k] = \varepsilon_{jkl} E_l$.

Proof. Using $E_j = -\frac{i}{2}\sigma_j$ and Lemma 2.10,

$$[E_j, E_k] = \left(-\frac{i}{2}\right)^2 [\sigma_j, \sigma_k] = -\frac{1}{4} \left((\delta_{jk}I + i\varepsilon_{jkl}\sigma_l) - (\delta_{kj}I + i\varepsilon_{kjl}\sigma_l) \right).$$

The δ terms cancel, and $\varepsilon_{kjl} = -\varepsilon_{jkl}$, so the bracket of the Pauli matrices is $[\sigma_j, \sigma_k] = 2i\varepsilon_{jkl}\sigma_l$. Hence

$$[E_j, E_k] = -\frac{1}{4} \cdot 2i\varepsilon_{jkl}\sigma_l = -\frac{i}{2}\varepsilon_{jkl}\sigma_l = \varepsilon_{jkl}E_l. \quad \square$$

Theorem 2.12 (Lie algebra isomorphism). *The linear maps determined by $\mathbf{e}_j \mapsto L_j$ and $\mathbf{e}_j \mapsto E_j$ give isomorphisms of real Lie algebras*

$$(\mathbb{R}^3, \times) \cong \mathfrak{so}(3) \cong \mathfrak{su}(2).$$

Proof. All three are 3-dimensional real vector spaces, and each map sends a basis to a basis, hence is a linear isomorphism. It remains only to check that brackets correspond, and they do because all three satisfy the *same* structure constants ε_{jkl} : $\mathbf{e}_j \times \mathbf{e}_k = \varepsilon_{jkl}\mathbf{e}_l$ by definition of the cross product, $[L_j, L_k] = \varepsilon_{jkl}L_l$ by Proposition 2.5, and $[E_j, E_k] = \varepsilon_{jkl}E_l$ by Proposition 2.11. $\square$

Remark 2.13. Theorem 2.12 is a statement about the *infinitesimal* objects only. The groups $\mathrm{SO}(3)$ and $\mathrm{SU}(2)$ are *not* isomorphic — Section 4 shows the map between them is two-to-one. Isomorphic Lie algebras with non-isomorphic groups is exactly the phenomenon that produces the half-angle and the sign flip that engineers meet in the field.

2.3 Quaternions

Definition 2.14. $\mathbb{H} = \{q_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k} : q_\mu \in \mathbb{R}\}$ with multiplication determined by $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1$ and $\mathbb{R}$ -bilinearity. The conjugate is $q^* = q_0 - q_1\mathbf{i} - q_2\mathbf{j} - q_3\mathbf{k}$ and the norm is $|q|^2 = q q^* = q_0^2 + q_1^2 + q_2^2 + q_3^2$.

Lemma 2.15 (Product in vector form). *For $p = (p_0, \mathbf{p})$ and $q = (q_0, \mathbf{q})$,*

$$p \otimes q = (p_0q_0 - \mathbf{p} \cdot \mathbf{q}, \quad p_0\mathbf{q} + q_0\mathbf{p} + \mathbf{p} \times \mathbf{q}).$$

Proof. First derive the products of units from the defining relations. From $\mathbf{ijk} = -1$, multiply on the right by $\mathbf{k}$: $\mathbf{ijk}^2 = -\mathbf{k}$, and since $\mathbf{k}^2 = -1$ this gives $-\mathbf{ij} = -\mathbf{k}$, i.e. $\mathbf{ij} = \mathbf{k}$. Multiplying $\mathbf{ij} = \mathbf{k}$ on the left by $\mathbf{i}$ gives $\mathbf{i}^2\mathbf{j} = \mathbf{ik}$, so $-\mathbf{j} = \mathbf{ik}$, i.e. $\mathbf{ki} = \mathbf{j}$ after a similar right multiplication. Cycling gives

$$\mathbf{ij} = \mathbf{k} = -\mathbf{ji}, \quad \mathbf{jk} = \mathbf{i} = -\mathbf{kj}, \quad \mathbf{ki} = \mathbf{j} = -\mathbf{ik}.$$

In one formula, with $\mathbf{e}_1 = \mathbf{i}$, $\mathbf{e}_2 = \mathbf{j}$, $\mathbf{e}_3 = \mathbf{k}$:

$$\mathbf{e}_j\mathbf{e}_k = -\delta_{jk} + \varepsilon_{jkl}\mathbf{e}_l.$$

Now expand the product bilinearly:

$$p \otimes q = (p_0 + p_j\mathbf{e}_j)(q_0 + q_k\mathbf{e}_k) = p_0q_0 + p_0q_k\mathbf{e}_k + q_0p_j\mathbf{e}_j + p_jq_k\mathbf{e}_j\mathbf{e}_k.$$

Substituting $\mathbf{e}_j\mathbf{e}_k = -\delta_{jk} + \varepsilon_{jkl}\mathbf{e}_l$ splits the last term into a scalar piece $-p_jq_j = -\mathbf{p} \cdot \mathbf{q}$ and a vector piece $\varepsilon_{jkl}p_jq_k\mathbf{e}_l = (\mathbf{p} \times \mathbf{q})_l\mathbf{e}_l$. Collecting scalar and vector parts gives the stated formula. $\square$

Proposition 2.16. *The map*

$$\Psi(q_0, \mathbf{q}) = q_0 I - \mathbf{i}(\mathbf{q} \cdot \boldsymbol{\sigma}) = \begin{pmatrix} q_0 - \mathbf{i}q_3 & -q_2 - \mathbf{i}q_1 \\ q_2 - \mathbf{i}q_1 & q_0 + \mathbf{i}q_3 \end{pmatrix}$$

restricts to a group isomorphism from the unit quaternions $(\mathbb{S}^3, \otimes)$ onto $\mathrm{SU}(2)$.

Proof. *Lands in $\mathrm{SU}(2)$.* Since each σ_j is Hermitian, $\Psi(q)^\dagger = q_0 I + \mathbf{i}(\mathbf{q} \cdot \boldsymbol{\sigma})$, so by Lemma 2.10

$$\Psi(q)^\dagger \Psi(q) = (q_0 I + \mathbf{i}\mathbf{q} \cdot \boldsymbol{\sigma})(q_0 I - \mathbf{i}\mathbf{q} \cdot \boldsymbol{\sigma}) = q_0^2 I + (\mathbf{q} \cdot \boldsymbol{\sigma})^2 = q_0^2 I + |\mathbf{q}|^2 I = |\mathbf{q}|^2 I,$$

using $(\mathbf{q} \cdot \boldsymbol{\sigma})^2 = (\mathbf{q} \cdot \mathbf{q})I + \mathbf{i}(\mathbf{q} \times \mathbf{q}) \cdot \boldsymbol{\sigma} = |\mathbf{q}|^2 I$ because $\mathbf{q} \times \mathbf{q} = \mathbf{0}$; note also the cross terms $\mp \mathbf{i}q_0(\mathbf{q} \cdot \boldsymbol{\sigma})$ cancel. Hence $|\mathbf{q}| = 1$ implies $\Psi(q)^\dagger \Psi(q) = I$. The determinant of the displayed matrix is

$$(q_0 - \mathbf{i}q_3)(q_0 + \mathbf{i}q_3) - (-q_2 - \mathbf{i}q_1)(q_2 - \mathbf{i}q_1) = (q_0^2 + q_3^2) + (q_2 + \mathbf{i}q_1)(q_2 - \mathbf{i}q_1) = q_0^2 + q_3^2 + q_2^2 + q_1^2 = 1.$$

Homomorphism. Using Lemma 2.10 again,

$$\begin{aligned} \Psi(p)\Psi(q) &= (p_0 I - \mathbf{i}\mathbf{p} \cdot \boldsymbol{\sigma})(q_0 I - \mathbf{i}\mathbf{q} \cdot \boldsymbol{\sigma}) \\ &= p_0 q_0 I - \mathbf{i}p_0(\mathbf{q} \cdot \boldsymbol{\sigma}) - \mathbf{i}q_0(\mathbf{p} \cdot \boldsymbol{\sigma}) - (\mathbf{p} \cdot \boldsymbol{\sigma})(\mathbf{q} \cdot \boldsymbol{\sigma}) \\ &= p_0 q_0 I - \mathbf{i}p_0(\mathbf{q} \cdot \boldsymbol{\sigma}) - \mathbf{i}q_0(\mathbf{p} \cdot \boldsymbol{\sigma}) - (\mathbf{p} \cdot \mathbf{q})I - \mathbf{i}(\mathbf{p} \times \mathbf{q}) \cdot \boldsymbol{\sigma} \\ &= (p_0 q_0 - \mathbf{p} \cdot \mathbf{q})I - \mathbf{i}(p_0 \mathbf{q} + q_0 \mathbf{p} + \mathbf{p} \times \mathbf{q}) \cdot \boldsymbol{\sigma}, \end{aligned}$$

which is exactly $\Psi(p \otimes q)$ by Lemma 2.15.

Bijectivity. Comparing with Proposition 2.8, setting $\alpha = q_0 - \mathbf{i}q_3$ and $\beta = q_2 - \mathbf{i}q_1$ recovers the general element of $\mathrm{SU}(2)$, and the correspondence $(q_0, q_1, q_2, q_3) \leftrightarrow (\alpha, \beta)$ is an $\mathbb{R}$ -linear bijection $\mathbb{R}^4 \rightarrow \mathbb{C}^2$ carrying $\mathbb{S}^3$ onto $\{|\alpha|^2 + |\beta|^2 = 1\}$. $\square$

Engineering translation: a unit quaternion is an $\mathrm{SU}(2)$ matrix

Nothing is lost or gained by storing attitude as four real numbers rather than a complex 2×2 matrix; Proposition 2.16 says they are the same object. The four-number form wins in flight software only for storage and bandwidth reasons: 4 doubles versus 8, and no complex arithmetic in the inner loop. When a physicist writes $\mathrm{SU}(2)$ and an engineer writes `Quaterniond`, they are pointing at the same group element.

3 The exponential maps, summed explicitly

3.1 $\exp$ on $\mathfrak{so}(3)$: Rodrigues' formula

Lemma 3.1. *Let $\mathbf{n} \in \mathbb{R}^3$ with $|\mathbf{n}| = 1$ and $K = \widehat{\mathbf{n}}$. Then*

$$K^2 = \mathbf{n}\mathbf{n}^\top - I, \quad K^3 = -K.$$

Proof. Apply both sides of the first claim to an arbitrary $\mathbf{w}$. By Lemma 2.4, $K^2 \mathbf{w} = \mathbf{n} \times (\mathbf{n} \times \mathbf{w})$. Use the standard triple product expansion $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$, proved in Appendix A:

$$K^2 \mathbf{w} = \mathbf{n}(\mathbf{n} \cdot \mathbf{w}) - \mathbf{w}(\mathbf{n} \cdot \mathbf{n}) = \mathbf{n}(\mathbf{n}^\top \mathbf{w}) - \mathbf{w} = (\mathbf{n}\mathbf{n}^\top - I)\mathbf{w},$$

using $|\mathbf{n}| = 1$. Since $\mathbf{w}$ was arbitrary, $K^2 = \mathbf{n}\mathbf{n}^\top - I$.

For the second claim,

$$K^3 = K \cdot K^2 = K(\mathbf{n}\mathbf{n}^\top - I) = (K\mathbf{n})\mathbf{n}^\top - K = \mathbf{0} \cdot \mathbf{n}^\top - K = -K,$$

because $K\mathbf{n} = \mathbf{n} \times \mathbf{n} = \mathbf{0}$. $\square$

Theorem 3.2 (Rodrigues). For $|\mathbf{n}| = 1$, $\theta \in \mathbb{R}$, and $K = \widehat{\mathbf{n}}$,

$$e^{\theta K} = I + \sin \theta K + (1 - \cos \theta) K^2$$

Proof. Because $K^3 = -K$, every power of K reduces to K or K^2 . By induction, for $m \geq 0$

$$K^{2m+1} = (-1)^m K, \quad K^{2m+2} = (-1)^m K^2.$$

Base: $m = 0$ gives $K^1 = K$ and $K^2 = K^2$. *Step:* assuming both hold at m ,

$$\begin{aligned} K^{2m+3} &= K^{2m+1} K^2 = (-1)^m K \cdot K^2 = (-1)^m K^3 = (-1)^{m+1} K, \\ K^{2m+4} &= K^{2m+2} K^2 = (-1)^m K^2 K^2 = (-1)^m K K^3 = (-1)^m K(-K) = (-1)^{m+1} K^2. \end{aligned}$$

Now split the exponential series by parity, isolating the $m = 0$ term I :

$$e^{\theta K} = \sum_{p \geq 0} \frac{\theta^p}{p!} K^p = I + \sum_{m \geq 0} \frac{\theta^{2m+1}}{(2m+1)!} K^{2m+1} + \sum_{m \geq 0} \frac{\theta^{2m+2}}{(2m+2)!} K^{2m+2}.$$

Substituting the reduced powers,

$$e^{\theta K} = I + \left(\sum_{m \geq 0} \frac{(-1)^m \theta^{2m+1}}{(2m+1)!} \right) K + \left(\sum_{m \geq 0} \frac{(-1)^m \theta^{2m+2}}{(2m+2)!} \right) K^2.$$

The first bracket is the Taylor series of $\sin \theta$. For the second, re-index with $r = m + 1$:

$$\sum_{m \geq 0} \frac{(-1)^m \theta^{2m+2}}{(2m+2)!} = \sum_{r \geq 1} \frac{(-1)^{r-1} \theta^{2r}}{(2r)!} = - \sum_{r \geq 1} \frac{(-1)^r \theta^{2r}}{(2r)!} = -(\cos \theta - 1) = 1 - \cos \theta,$$

where the last step adds and subtracts the $r = 0$ term of the cosine series. This is the claim. $\square$

Corollary 3.3. $e^{\theta \widehat{\mathbf{n}}} \mathbf{v} = \cos \theta \mathbf{v} + \sin \theta (\mathbf{n} \times \mathbf{v}) + (1 - \cos \theta)(\mathbf{n} \cdot \mathbf{v}) \mathbf{n}$.

Proof. Apply Theorem 3.2 to $\mathbf{v}$ and use $K\mathbf{v} = \mathbf{n} \times \mathbf{v}$ and $K^2\mathbf{v} = (\mathbf{n}\mathbf{n}^\top - I)\mathbf{v} = (\mathbf{n} \cdot \mathbf{v})\mathbf{n} - \mathbf{v}$:

$$e^{\theta K} \mathbf{v} = \mathbf{v} + \sin \theta (\mathbf{n} \times \mathbf{v}) + (1 - \cos \theta) [(\mathbf{n} \cdot \mathbf{v}) \mathbf{n} - \mathbf{v}] = \cos \theta \mathbf{v} + \sin \theta (\mathbf{n} \times \mathbf{v}) + (1 - \cos \theta)(\mathbf{n} \cdot \mathbf{v}) \mathbf{n}. \quad \square$$

Engineering translation: the only three matrices you ever need to build a rotation

Theorem 3.2 says a rotation matrix is I , plus $\sin \theta$ times a cross-product matrix, plus $(1 - \cos \theta)$ times its square. No trigonometric bookkeeping of axes, no Euler sequence. In code this is three `axpy` operations on 3×3 blocks and two transcendental calls. If you are building R from an axis and angle any other way, you are doing more work and introducing more branches than the mathematics requires.

Pitfall: the $\theta \rightarrow 0$ cancellation

As $\theta \rightarrow 0$, $1 - \cos \theta \rightarrow 0$ like $\theta^2/2$, and computing it as a literal $1 - \cos \theta$ loses about half your significant digits by cancellation: at $\theta = 10^{-8}$, $\cos \theta$ rounds to 1 exactly in double precision and

you get 0 instead of 5×10^{-17} . Use the algebraically equivalent forms

$$1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}, \quad \frac{\sin \theta}{\theta}, \quad \frac{1 - \cos \theta}{\theta^2},$$

and switch to the truncated series $\frac{\sin \theta}{\theta} \approx 1 - \frac{\theta^2}{6}$, $\frac{1 - \cos \theta}{\theta^2} \approx \frac{1}{2} - \frac{\theta^2}{24}$ below a threshold of roughly $\theta < 10^{-4}$. This single change is the difference between an attitude propagator that holds orthogonality for a whole mission and one that drifts visibly in an hour.

3.2 exp on $\mathfrak{su}(2)$: where the half-angle comes from

Theorem 3.4. For $|\mathbf{n}| = 1$ and $\theta \in \mathbb{R}$,

$$\exp(\theta \mathbf{n} \cdot \mathbf{E}) = \exp\left(-\frac{i\theta}{2} \mathbf{n} \cdot \boldsymbol{\sigma}\right) = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (\mathbf{n} \cdot \boldsymbol{\sigma})$$

where $\mathbf{n} \cdot \mathbf{E} = n_j E_j$.

Proof. Write $S = \mathbf{n} \cdot \boldsymbol{\sigma}$. By Lemma 2.10,

$$S^2 = (\mathbf{n} \cdot \boldsymbol{\sigma})(\mathbf{n} \cdot \boldsymbol{\sigma}) = (\mathbf{n} \cdot \mathbf{n})I + i(\mathbf{n} \times \mathbf{n}) \cdot \boldsymbol{\sigma} = I,$$

since $\mathbf{n} \times \mathbf{n} = \mathbf{0}$ and $|\mathbf{n}| = 1$. Hence $S^{2m} = I$ and $S^{2m+1} = S$ for all $m \geq 0$. Now expand with $\phi := \theta/2$:

$$e^{-i\phi S} = \sum_{p \geq 0} \frac{(-i\phi)^p}{p!} S^p = \sum_{m \geq 0} \frac{(-i\phi)^{2m}}{(2m)!} \underbrace{S^{2m}}_{=I} + \sum_{m \geq 0} \frac{(-i\phi)^{2m+1}}{(2m+1)!} \underbrace{S^{2m+1}}_{=S}.$$

For the even sum, $(-i\phi)^{2m} = ((-i)^2)^m \phi^{2m} = (-1)^m \phi^{2m}$, so

$$\sum_{m \geq 0} \frac{(-i\phi)^{2m}}{(2m)!} I = \left(\sum_{m \geq 0} \frac{(-1)^m \phi^{2m}}{(2m)!} \right) I = \cos \phi I.$$

For the odd sum, $(-i\phi)^{2m+1} = (-1)^m (-i) \phi^{2m+1}$, so

$$\sum_{m \geq 0} \frac{(-i\phi)^{2m+1}}{(2m+1)!} S = -i \left(\sum_{m \geq 0} \frac{(-1)^m \phi^{2m+1}}{(2m+1)!} \right) S = -i \sin \phi S.$$

Adding the two gives $e^{-i\phi S} = \cos \phi I - i \sin \phi S$ with $\phi = \theta/2$. □

Corollary 3.5 (Axis-angle to quaternion). Under Ψ of Proposition 2.16, the group element $\exp(\theta \mathbf{n} \cdot \mathbf{E})$ corresponds to the unit quaternion

$$q = \left(\cos \frac{\theta}{2}, \sin \frac{\theta}{2} \mathbf{n} \right).$$

Proof. $\Psi(q_0, \mathbf{q}) = q_0 I - i \mathbf{q} \cdot \boldsymbol{\sigma}$. Matching against $\cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (\mathbf{n} \cdot \boldsymbol{\sigma})$ gives $q_0 = \cos \frac{\theta}{2}$ and $\mathbf{q} = \sin \frac{\theta}{2} \mathbf{n}$. □

Remark 3.6 (The half-angle is not a convention). Compare Theorem 3.2 and Theorem 3.4: the same Lie algebra element $\theta \mathbf{n}$, exponentiated in the two groups, produces θ in $\text{SO}(3)$ and $\theta/2$ in $\text{SU}(2)$. The factor of two is forced by the identification of Lie algebras in Theorem 2.12 together with the two-to-one group map of Section 4: the $\text{SU}(2)$ parameter must run twice as slowly so that a 2π rotation in $\text{SO}(3)$ corresponds to a 4π traversal in $\text{SU}(2)$.

4 The covering homomorphism $\mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$

Definition 4.1. For $\mathbf{v} \in \mathbb{R}^3$ put $X(\mathbf{v}) = \mathbf{v} \cdot \mathbf{E} = -\frac{i}{2}(\mathbf{v} \cdot \boldsymbol{\sigma}) \in \mathfrak{su}(2)$, a real-linear isomorphism $\mathbb{R}^3 \rightarrow \mathfrak{su}(2)$. For $U \in \mathrm{SU}(2)$ define the adjoint action $\mathrm{Ad}_U : \mathfrak{su}(2) \rightarrow \mathfrak{su}(2)$, $\mathrm{Ad}_U(X) = UXU^\dagger$, and define $\varphi(U) \in \mathbb{R}^{3 \times 3}$ by

$$X(\varphi(U)\mathbf{v}) = UX(\mathbf{v})U^\dagger.$$

Lemma 4.2. Ad_U maps $\mathfrak{su}(2)$ into $\mathfrak{su}(2)$, so $\varphi(U)$ is well defined and linear.

Proof. Let $X \in \mathfrak{su}(2)$, so $X^\dagger = -X$ and $\mathrm{tr} X = 0$. Then $(UXU^\dagger)^\dagger = UX^\dagger U^\dagger = -UXU^\dagger$, so $UXU^\dagger$ is anti-Hermitian; and $\mathrm{tr}(UXU^\dagger) = \mathrm{tr}(XU^\dagger U) = \mathrm{tr} X = 0$ by cyclicity of the trace and $U^\dagger U = I$. Linearity of $\varphi(U)$ follows because X is linear and $X \mapsto UXU^\dagger$ is linear. $\square$

Lemma 4.3 (The invariant inner product). For all $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$, $-2 \mathrm{tr}(X(\mathbf{v})X(\mathbf{w})) = \mathbf{v} \cdot \mathbf{w}$.

Proof. By Lemma 2.10,

$$X(\mathbf{v})X(\mathbf{w}) = \left(-\frac{i}{2}\right)^2 (\mathbf{v} \cdot \boldsymbol{\sigma})(\mathbf{w} \cdot \boldsymbol{\sigma}) = -\frac{1}{4}[(\mathbf{v} \cdot \mathbf{w})I + i(\mathbf{v} \times \mathbf{w}) \cdot \boldsymbol{\sigma}].$$

Now $\mathrm{tr} I = 2$ and $\mathrm{tr} \sigma_l = 0$ for each l (immediate from the explicit matrices), so

$$\mathrm{tr}(X(\mathbf{v})X(\mathbf{w})) = -\frac{1}{4}[(\mathbf{v} \cdot \mathbf{w}) \cdot 2 + 0] = -\frac{1}{2} \mathbf{v} \cdot \mathbf{w}.$$

Multiplying by -2 gives the claim. $\square$

Theorem 4.4. $\varphi : \mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$ is a surjective group homomorphism with kernel $\{+I, -I\}$. Hence

$$\mathrm{SO}(3) \cong \mathrm{SU}(2)/\{\pm I\} \cong \mathbb{S}^3/\sim = \mathbb{RP}^3.$$

Proof. *Orthogonality.* Using Lemma 4.3 twice and cyclicity of the trace,

$$\begin{aligned} \varphi(U)\mathbf{v} \cdot \varphi(U)\mathbf{w} &= -2 \mathrm{tr}(X(\varphi(U)\mathbf{v})X(\varphi(U)\mathbf{w})) = -2 \mathrm{tr}(UX(\mathbf{v})U^\dagger UX(\mathbf{w})U^\dagger) \\ &= -2 \mathrm{tr}(UX(\mathbf{v})X(\mathbf{w})U^\dagger) = -2 \mathrm{tr}(X(\mathbf{v})X(\mathbf{w})U^\dagger U) = -2 \mathrm{tr}(X(\mathbf{v})X(\mathbf{w})) = \mathbf{v} \cdot \mathbf{w}. \end{aligned}$$

So $\varphi(U)$ preserves the inner product, i.e. $\varphi(U) \in \mathrm{O}(3)$.

Determinant +1. $\mathrm{SU}(2) \cong \mathbb{S}^3$ is path connected, and φ is continuous (it is polynomial in the entries of U). Therefore $\det \varphi(U)$ is a continuous function on a connected space taking values in the discrete set $\{\pm 1\}$, hence constant. At $U = I$ we get $\varphi(I) = I$ and $\det = +1$, so $\det \varphi(U) = +1$ for all U .

Homomorphism. For $U, V \in \mathrm{SU}(2)$,

$$X(\varphi(UV)\mathbf{v}) = (UV)X(\mathbf{v})(UV)^\dagger = U(VX(\mathbf{v})V^\dagger)U^\dagger = UX(\varphi(V)\mathbf{v})U^\dagger = X(\varphi(U)\varphi(V)\mathbf{v}),$$

and since X is injective, $\varphi(UV) = \varphi(U)\varphi(V)$.

Kernel. Suppose $\varphi(U) = I$, i.e. $UX(\mathbf{v})U^\dagger = X(\mathbf{v})$ for every $\mathbf{v}$, equivalently $U\sigma_j U^\dagger = \sigma_j$, i.e. U commutes with $\sigma_1, \sigma_2, \sigma_3$. Rather than invoke Schur's lemma, compute. Write $U = \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix}$. Commuting with $\sigma_3 = \mathrm{diag}(1, -1)$:

$$U\sigma_3 = \begin{pmatrix} \alpha & \bar{\beta} \\ \beta & -\bar{\alpha} \end{pmatrix}, \quad \sigma_3 U = \begin{pmatrix} \alpha & -\bar{\beta} \\ -\beta & \bar{\alpha} \end{pmatrix}.$$

Equality of the (1, 2) entries forces $\bar{\beta} = -\beta$, so $\beta = 0$; then $|\alpha|^2 = 1$ and $U = \text{diag}(\alpha, \bar{\alpha})$. Commuting this with σ_1 :

$$U\sigma_1 = \begin{pmatrix} 0 & \alpha \\ \bar{\alpha} & 0 \end{pmatrix}, \quad \sigma_1 U = \begin{pmatrix} 0 & \bar{\alpha} \\ \alpha & 0 \end{pmatrix},$$

so $\alpha = \bar{\alpha}$, i.e. $\alpha \in \mathbb{R}$, and with $|\alpha| = 1$ this gives $\alpha = \pm 1$, that is $U = \pm I$. Conversely $\pm I$ clearly act trivially since $(-I)X(-I)^\dagger = X$.

Surjectivity. Let $R \in \text{SO}(3)$. By Corollary 4.5 below, φ maps $\exp(\theta \mathbf{n} \cdot \mathbf{E})$ to $\exp(\theta \hat{\mathbf{n}})$, and every element of $\text{SO}(3)$ is of the latter form by the axis-angle theorem (a real orthogonal 3×3 matrix with $\det = +1$ has $+1$ as an eigenvalue, whose eigenvector is the axis; restricted to the orthogonal complement it is a planar rotation). Hence φ is onto.

The isomorphism statements are the first isomorphism theorem plus the observation that quotienting $\mathbb{S}^3$ by the antipodal identification $q \sim -q$ is precisely the definition of $\mathbb{RP}^3$. $\square$

Corollary 4.5. $\varphi(\exp(\theta \mathbf{n} \cdot \mathbf{E})) = \exp(\theta \hat{\mathbf{n}})$; equivalently $d\varphi_I = \text{id}$ under the identifications $\mathbf{e}_j \mapsto E_j$ and $\mathbf{e}_j \mapsto L_j$.

Proof. Differentiate the defining relation $X(\varphi(U)\mathbf{v}) = UX(\mathbf{v})U^\dagger$ along $U(t) = \exp(tY)$ with $Y \in \mathfrak{su}(2)$, at $t = 0$:

$$X(d\varphi_I(Y)\mathbf{v}) = YX(\mathbf{v}) - X(\mathbf{v})Y = [Y, X(\mathbf{v})].$$

Take $Y = E_j$ and $\mathbf{v} = \mathbf{e}_k$, so $X(\mathbf{v}) = E_k$. By Proposition 2.11, $[E_j, E_k] = \varepsilon_{jkl}E_l = X(\varepsilon_{jkl}\mathbf{e}_l) = X(\mathbf{e}_j \times \mathbf{e}_k) = X(\hat{\mathbf{e}}_j \mathbf{e}_k)$. Injectivity of X gives $d\varphi_I(E_j) = \hat{\mathbf{e}}_j = L_j$. Since $d\varphi_I$ is a Lie algebra homomorphism sending $E_j \mapsto L_j$, naturality of $\exp$ ($\varphi \circ \exp = \exp \circ d\varphi_I$ for a Lie group homomorphism φ) yields the displayed identity. $\square$

Engineering translation: why your quaternion flips sign, and what to do about it

Theorem 4.4 is the entire explanation. q and $-q$ are distinct points of $\mathbb{S}^3$ but map to the *same* rotation, because the kernel of φ is $\{\pm I\}$. Consequences you will meet:

- A filter or a solver may output $-q$ where you expected q . Nothing is wrong.
- Comparing attitudes with $\|q_1 - q_2\|$ is **incorrect**. Use $1 - |\langle q_1, q_2 \rangle|$, or the geodesic angle $2 \arccos |\langle q_1, q_2 \rangle|$ — the absolute value is what quotients by the sign.
- Before interpolating between q_1 and q_2 , negate one if $\langle q_1, q_2 \rangle < 0$; otherwise you take the long way round the sphere ($> 180^\circ$).
- Canonicalizing to $q_0 \geq 0$ is fine for logging and comparison, but never do it *inside* an integrator: it injects a discontinuity into a signal your propagator assumes is smooth.

Remark 4.6 (The belt trick is a statement about π_1). $\mathbb{S}^3$ is simply connected and the covering is two-sheeted, so $\pi_1(\text{SO}(3)) \cong \ker \varphi \cong \mathbb{Z}/2\mathbb{Z}$. Physically: a loop of rotations through 2π is not contractible, while a loop through 4π is. This is the belt or plate trick, and it is why $\text{spin-}\frac{1}{2}$ exists. It is also why a naive “unwind the total rotation angle” bookkeeping in software must decide, explicitly, whether it is tracking an element of $\text{SO}(3)$ or a path in $\text{SU}(2)$.

5 Quaternion to rotation matrix, every step

Theorem 5.1. *Let $q = (q_0, \mathbf{q})$ be a unit quaternion and $\mathbf{v} \in \mathbb{R}^3$ embedded as the pure quaternion $(0, \mathbf{v})$. Then $q \otimes (0, \mathbf{v}) \otimes q^*$ is again pure, and its vector part is*

$$\mathbf{v}' = (q_0^2 - |\mathbf{q}|^2)\mathbf{v} + 2(\mathbf{q} \cdot \mathbf{v})\mathbf{q} + 2q_0(\mathbf{q} \times \mathbf{v})$$

Proof. Step 1: the inner product $q \otimes (0, \mathbf{v})$. By Lemma 2.15 with $p = (q_0, \mathbf{q})$ and the second factor $(0, \mathbf{v})$,

$$q \otimes (0, \mathbf{v}) = (q_0 \cdot 0 - \mathbf{q} \cdot \mathbf{v}, \quad q_0\mathbf{v} + 0 \cdot \mathbf{q} + \mathbf{q} \times \mathbf{v}) = (-\mathbf{q} \cdot \mathbf{v}, \quad q_0\mathbf{v} + \mathbf{q} \times \mathbf{v}).$$

Step 2: multiply on the right by $q^ = (q_0, -\mathbf{q})$.* Apply Lemma 2.15 with $p = (-\mathbf{q} \cdot \mathbf{v}, \quad q_0\mathbf{v} + \mathbf{q} \times \mathbf{v})$ and $q^* = (q_0, -\mathbf{q})$.

The scalar part is

$$(-\mathbf{q} \cdot \mathbf{v})q_0 - (q_0\mathbf{v} + \mathbf{q} \times \mathbf{v}) \cdot (-\mathbf{q}) = -q_0(\mathbf{q} \cdot \mathbf{v}) + q_0(\mathbf{v} \cdot \mathbf{q}) + (\mathbf{q} \times \mathbf{v}) \cdot \mathbf{q}.$$

The first two terms cancel exactly. The third vanishes because $\mathbf{q} \times \mathbf{v}$ is orthogonal to $\mathbf{q}$: indeed $(\mathbf{q} \times \mathbf{v}) \cdot \mathbf{q} = \varepsilon_{jkl}q_k v_l q_j = 0$, since ε_{jkl} is antisymmetric in $j \leftrightarrow k$ while $q_j q_k$ is symmetric, and the contraction of a symmetric with an antisymmetric tensor is zero. So the result is pure, as claimed.

The vector part is, term by term,

$$\underbrace{(-\mathbf{q} \cdot \mathbf{v})(-\mathbf{q})}_{\text{first scalar} \times \text{second vector}} + \underbrace{q_0(q_0\mathbf{v} + \mathbf{q} \times \mathbf{v})}_{\text{second scalar} \times \text{first vector}} + \underbrace{(q_0\mathbf{v} + \mathbf{q} \times \mathbf{v}) \times (-\mathbf{q})}_{\text{cross term}}.$$

Handle each piece.

- First piece: $(-\mathbf{q} \cdot \mathbf{v})(-\mathbf{q}) = (\mathbf{q} \cdot \mathbf{v})\mathbf{q}$.
- Second piece: $q_0^2\mathbf{v} + q_0(\mathbf{q} \times \mathbf{v})$.
- Third piece: expand the cross product and use antisymmetry $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$:

$$(q_0\mathbf{v}) \times (-\mathbf{q}) + (\mathbf{q} \times \mathbf{v}) \times (-\mathbf{q}) = -q_0(\mathbf{v} \times \mathbf{q}) - (\mathbf{q} \times \mathbf{v}) \times \mathbf{q} = q_0(\mathbf{q} \times \mathbf{v}) + \mathbf{q} \times (\mathbf{q} \times \mathbf{v}).$$

Now apply the triple product expansion (Appendix A): $\mathbf{q} \times (\mathbf{q} \times \mathbf{v}) = \mathbf{q}(\mathbf{q} \cdot \mathbf{v}) - \mathbf{v}(\mathbf{q} \cdot \mathbf{q}) = (\mathbf{q} \cdot \mathbf{v})\mathbf{q} - |\mathbf{q}|^2\mathbf{v}$.

Summing the three pieces:

$$\mathbf{v}' = (\mathbf{q} \cdot \mathbf{v})\mathbf{q} + q_0^2\mathbf{v} + q_0(\mathbf{q} \times \mathbf{v}) + q_0(\mathbf{q} \times \mathbf{v}) + (\mathbf{q} \cdot \mathbf{v})\mathbf{q} - |\mathbf{q}|^2\mathbf{v}.$$

Collecting like terms — two copies of $(\mathbf{q} \cdot \mathbf{v})\mathbf{q}$, two copies of $q_0(\mathbf{q} \times \mathbf{v})$, and the $\mathbf{v}$ terms — gives exactly the boxed formula. $\square$

Corollary 5.2 (Consistency with Rodrigues). *Substituting $q_0 = \cos \frac{\theta}{2}$ and $\mathbf{q} = \sin \frac{\theta}{2} \mathbf{n}$ into Theorem 5.1 reproduces Corollary 3.3.*

Proof. Coefficient of $\mathbf{v}$: $\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} = \cos \theta$ by the double-angle identity. Coefficient of $(\mathbf{n} \cdot \mathbf{v})\mathbf{n}$: $2 \sin^2 \frac{\theta}{2} = 1 - \cos \theta$, again double angle. Coefficient of $\mathbf{n} \times \mathbf{v}$: $2 \cos \frac{\theta}{2} \sin \frac{\theta}{2} = \sin \theta$. Substituting,

$$\mathbf{v}' = \cos \theta \mathbf{v} + (1 - \cos \theta)(\mathbf{n} \cdot \mathbf{v})\mathbf{n} + \sin \theta (\mathbf{n} \times \mathbf{v}),$$

which is Corollary 3.3. This also confirms independently that the half-angle in Corollary 3.5 is correct: a quaternion built with $\theta/2$ produces a rotation by θ . $\square$

Corollary 5.3 (Matrix form). *With q a unit quaternion,*

$$R(q) = (q_0^2 - |\mathbf{q}|^2)I + 2\mathbf{q}\mathbf{q}^\top + 2q_0\widehat{\mathbf{q}} = \begin{pmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 - q_0q_3) & 2(q_1q_3 + q_0q_2) \\ 2(q_1q_2 + q_0q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2q_3 - q_0q_1) \\ 2(q_1q_3 - q_0q_2) & 2(q_2q_3 + q_0q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{pmatrix}.$$

Proof. The three terms of Theorem 5.1 are, as operators on $\mathbf{v}$: multiplication by the scalar $q_0^2 - |\mathbf{q}|^2$; the map $\mathbf{v} \mapsto 2(\mathbf{q} \cdot \mathbf{v})\mathbf{q} = 2\mathbf{q}\mathbf{q}^\top \mathbf{v}$; and $\mathbf{v} \mapsto 2q_0(\mathbf{q} \times \mathbf{v}) = 2q_0\widehat{\mathbf{q}}\mathbf{v}$ by Lemma 2.4. For the entries, take the (1,1) element: $q_0^2 - |\mathbf{q}|^2 + 2q_1^2 + 0 = q_0^2 - q_1^2 - q_2^2 - q_3^2 + 2q_1^2 = q_0^2 + q_1^2 - q_2^2 - q_3^2$. Take the (1,2) element: the diagonal term contributes nothing off-diagonal, $2\mathbf{q}\mathbf{q}^\top$ contributes $2q_1q_2$, and $2q_0\widehat{\mathbf{q}}$ contributes $2q_0 \cdot (-q_3)$, giving $2(q_1q_2 - q_0q_3)$. The remaining entries follow identically. $\square$

6 Rotation matrix to quaternion: Shepperd's method, derived

This is the direction that breaks in production code, because the naive formula divides by a quantity that can vanish.

Lemma 6.1 (Trace and antisymmetric part). *For $R = R(q)$ as in Corollary 5.3 with $|q| = 1$,*

$$\text{tr } R = 4q_0^2 - 1, \quad R - R^\top = 4q_0\widehat{\mathbf{q}}.$$

Proof. Trace: sum the three diagonal entries from Corollary 5.3,

$$\text{tr } R = (q_0^2 + q_1^2 - q_2^2 - q_3^2) + (q_0^2 - q_1^2 + q_2^2 - q_3^2) + (q_0^2 - q_1^2 - q_2^2 + q_3^2) = 3q_0^2 - q_1^2 - q_2^2 - q_3^2.$$

Using $q_1^2 + q_2^2 + q_3^2 = 1 - q_0^2$, this is $3q_0^2 - (1 - q_0^2) = 4q_0^2 - 1$.

Antisymmetric part: in $R = (q_0^2 - |\mathbf{q}|^2)I + 2\mathbf{q}\mathbf{q}^\top + 2q_0\widehat{\mathbf{q}}$ the first two terms are symmetric ($I^\top = I$ and $(\mathbf{q}\mathbf{q}^\top)^\top = \mathbf{q}\mathbf{q}^\top$) and the third is antisymmetric, so $R - R^\top = 2q_0\widehat{\mathbf{q}} - 2q_0\widehat{\mathbf{q}}^\top = 2q_0\widehat{\mathbf{q}} + 2q_0\widehat{\mathbf{q}} = 4q_0\widehat{\mathbf{q}}$. $\square$

Corollary 6.2 (Naive branch). *If $\text{tr } R > -1$ then $q_0 = \frac{1}{2}\sqrt{1 + \text{tr } R}$ (choosing the + sign) and*

$$q_1 = \frac{R_{32} - R_{23}}{4q_0}, \quad q_2 = \frac{R_{13} - R_{31}}{4q_0}, \quad q_3 = \frac{R_{21} - R_{12}}{4q_0}.$$

Proof. From $\text{tr } R = 4q_0^2 - 1$ we get $q_0^2 = (1 + \text{tr } R)/4$. From $R - R^\top = 4q_0\widehat{\mathbf{q}}$, read off the (3,2) entry: $(\widehat{\mathbf{q}})_{32} = q_1$, so $R_{32} - R_{23} = 4q_0q_1$. The (1,3) entry gives $(\widehat{\mathbf{q}})_{13} = q_2$, so $R_{13} - R_{31} = 4q_0q_2$; the (2,1) entry gives $(\widehat{\mathbf{q}})_{21} = q_3$, so $R_{21} - R_{12} = 4q_0q_3$. $\square$

Lemma 6.3 (The other three branches). *With $R = R(q)$, $|q| = 1$,*

$$1 + R_{11} - R_{22} - R_{33} = 4q_1^2, \quad 1 - R_{11} + R_{22} - R_{33} = 4q_2^2, \quad 1 - R_{11} - R_{22} + R_{33} = 4q_3^2.$$

Moreover $R_{12} + R_{21} = 4q_1q_2$, $R_{13} + R_{31} = 4q_1q_3$, $R_{23} + R_{32} = 4q_2q_3$.

Proof. For the first, substitute the diagonal entries of Corollary 5.3:

$$\begin{aligned} R_{11} - R_{22} - R_{33} &= (q_0^2 + q_1^2 - q_2^2 - q_3^2) - (q_0^2 - q_1^2 + q_2^2 - q_3^2) - (q_0^2 - q_1^2 - q_2^2 + q_3^2) \\ &= q_0^2 + q_1^2 - q_2^2 - q_3^2 - q_0^2 + q_1^2 - q_2^2 + q_3^2 - q_0^2 + q_1^2 + q_2^2 - q_3^2 \\ &= -q_0^2 + 3q_1^2 - q_2^2 - q_3^2. \end{aligned}$$

Adding $1 = q_0^2 + q_1^2 + q_2^2 + q_3^2$ gives $4q_1^2$, as claimed. The other two follow by permuting the roles of the indices 1, 2, 3, which permutes the diagonal entries in the same pattern.

For the symmetric off-diagonal identities, from Corollary 5.3 $R_{12} = 2(q_1q_2 - q_0q_3)$ and $R_{21} = 2(q_1q_2 + q_0q_3)$, so their sum is $4q_1q_2$; the antisymmetric q_0 pieces cancel. Identically $R_{13} + R_{31} = 2(q_1q_3 + q_0q_2)$ and $R_{23} + R_{32} = 2(q_2q_3 - q_0q_1) + 2(q_2q_3 + q_0q_1) = 4q_2q_3$. $\square$

Theorem 6.4 (Shepperd’s four-case algorithm). *Compute the four quantities*

$$t_0 = 1 + \text{tr } R = 4q_0^2, \quad t_1 = 1 + R_{11} - R_{22} - R_{33} = 4q_1^2,$$

$$t_2 = 1 - R_{11} + R_{22} - R_{33} = 4q_2^2, \quad t_3 = 1 - R_{11} - R_{22} + R_{33} = 4q_3^2,$$

select the index μ with t_μ largest, set $q_\mu = \frac{1}{2}\sqrt{t_\mu}$, and obtain the remaining components by dividing the appropriate identity of Lemma 6.1 or Lemma 6.3 by $4q_\mu$. Explicitly:

$$\begin{aligned} \mu = 0 : \quad q_1 &= \frac{R_{32} - R_{23}}{4q_0}, \quad q_2 = \frac{R_{13} - R_{31}}{4q_0}, \quad q_3 = \frac{R_{21} - R_{12}}{4q_0}; \\ \mu = 1 : \quad q_0 &= \frac{R_{32} - R_{23}}{4q_1}, \quad q_2 = \frac{R_{12} + R_{21}}{4q_1}, \quad q_3 = \frac{R_{13} + R_{31}}{4q_1}; \\ \mu = 2 : \quad q_0 &= \frac{R_{13} - R_{31}}{4q_2}, \quad q_1 = \frac{R_{12} + R_{21}}{4q_2}, \quad q_3 = \frac{R_{23} + R_{32}}{4q_2}; \\ \mu = 3 : \quad q_0 &= \frac{R_{21} - R_{12}}{4q_3}, \quad q_1 = \frac{R_{13} + R_{31}}{4q_3}, \quad q_2 = \frac{R_{23} + R_{32}}{4q_3}. \end{aligned}$$

The selected t_μ always satisfies $t_\mu \geq 1$, so the division is never by a small number.

Proof. The four displayed identities are Lemma 6.1 (for t_0) and Lemma 6.3 (for t_1, t_2, t_3). Each case then solves for the remaining three components using the products $4q_\mu q_\nu$ supplied by those two lemmas.

For the conditioning claim, sum the four quantities:

$$t_0 + t_1 + t_2 + t_3 = 4(q_0^2 + q_1^2 + q_2^2 + q_3^2) = 4.$$

Four non-negative numbers summing to 4 must have a maximum at least their average, namely $\max_\mu t_\mu \geq 1$. Hence $q_\mu = \frac{1}{2}\sqrt{t_\mu} \geq \frac{1}{2}$ and the divisor $4q_\mu \geq 2$. $\square$

Engineering translation: why the textbook one-liner is a latent bug

Corollary 6.2 is the formula printed in most references. It divides by $4q_0$, and $q_0 = \cos \frac{\theta}{2} \rightarrow 0$ as $\theta \rightarrow \pi$. A 180° rotation is not exotic — it is a spacecraft flipped end for end, a common commanded slew. Near it the naive formula loses precision catastrophically and at exactly $\theta = \pi$ it divides by zero.

Theorem 6.4 costs one extra comparison chain and is unconditionally well conditioned, because the pigeonhole argument guarantees the chosen divisor is at least 2. Use it always. When you review someone’s attitude library, “does the matrix-to-quaternion routine branch on four cases?” is a one-question audit of whether the author knew what they were doing.

Pitfall: feeding it a matrix that is not quite a rotation

Every identity above assumed $R^\top R = I$ exactly. After a long propagation, R drifts and t_μ can go slightly negative, making $\sqrt{t_\mu}$ a NaN that then poisons the entire state vector. Guard with $\max(t_\mu, 0)$, and re-orthogonalize R (polar decomposition, or Gram–Schmidt if you must) *before* conversion rather than debugging NaNs afterwards.

7 The logarithm and its singularities

Proposition 7.1. *Let $R \in \text{SO}(3)$ with $\text{tr } R \neq -1$ and $R \neq I$. Then $R = \exp(\theta \hat{\mathbf{n}})$ with*

$$\theta = \arccos\left(\frac{\text{tr } R - 1}{2}\right) \in (0, \pi), \quad \hat{\mathbf{n}} = \frac{R - R^\top}{2 \sin \theta}.$$

Proof. Take the trace of Rodrigues’ formula (Theorem 3.2). We have $\text{tr } I = 3$, $\text{tr } K = 0$ (a skew matrix has zero diagonal), and by Lemma 3.1 $\text{tr } K^2 = \text{tr}(\mathbf{n}\mathbf{n}^\top - I) = |\mathbf{n}|^2 - 3 = 1 - 3 = -2$. Hence

$$\text{tr } R = 3 + \sin \theta \cdot 0 + (1 - \cos \theta)(-2) = 3 - 2 + 2 \cos \theta = 1 + 2 \cos \theta,$$

which rearranges to $\cos \theta = (\text{tr } R - 1)/2$. For the axis, the symmetric terms I and K^2 drop out of $R - R^\top$, leaving $R - R^\top = 2 \sin \theta K$, so $K = (R - R^\top)/(2 \sin \theta)$ provided $\sin \theta \neq 0$, i.e. $\theta \notin \{0, \pi\}$, which is exactly the excluded set. $\square$

Engineering translation: the two singular attitudes, and the fix

The logarithm degenerates in precisely two places, and both are physically ordinary:

- $\theta \rightarrow 0$ ($R \rightarrow I$): the axis is undefined because there is no rotation. Any unit vector will do; return $\mathbf{0}$ for the rotation vector and do not normalize.
- $\theta \rightarrow \pi$ ($\text{tr } R \rightarrow -1$): $\sin \theta \rightarrow 0$ and the axis formula divides by zero, yet the axis is perfectly well defined. Recover it from the *symmetric* part instead: $R + R^\top = 2[(1 - \cos \theta)\mathbf{n}\mathbf{n}^\top + \cos \theta I]$, so at $\theta = \pi$, $\mathbf{n}\mathbf{n}^\top = (R + I)/2$, and $\mathbf{n}$ is the properly normalized column of $R + I$ with the largest norm. The sign of $\mathbf{n}$ is genuinely ambiguous here — rotating by $+\pi$ and $-\pi$ about the same axis give the same R — which is the double cover asserting itself again.

Go through the quaternion instead and both problems soften: build q by Theorem 6.4, then $\theta = 2 \text{atan2}(|\mathbf{q}|, q_0)$. Using `atan2` rather than `arccos` is not cosmetic: `arccos` has infinite derivative at its endpoints, so it amplifies input error precisely where you can least afford it, whereas `atan2` is well conditioned over the whole range.

8 Kinematics: the equations you actually integrate

Proposition 8.1. *Let $R(t) \in \text{SO}(3)$ be differentiable, carrying body coordinates to inertial coordinates. Then there exist unique $\boldsymbol{\omega}_B(t), \boldsymbol{\omega}_I(t) \in \mathbb{R}^3$ with*

$$\dot{R} = R \widehat{\boldsymbol{\omega}}_B = \widehat{\boldsymbol{\omega}}_I R, \quad \text{and} \quad \boldsymbol{\omega}_I = R \boldsymbol{\omega}_B.$$

Proof. Differentiate $R^\top R = I$:

$$\dot{R}^\top R + R^\top \dot{R} = 0 \implies (R^\top \dot{R})^\top = -(R^\top \dot{R}),$$

so $R^\top \dot{R}$ is skew-symmetric and by Proposition 2.2 equals $\widehat{\omega}_B$ for a unique ω_B . Multiplying on the left by R gives $\dot{R} = R\widehat{\omega}_B$. Similarly differentiating $RR^\top = I$ shows $\dot{R}R^\top$ is skew, $= \widehat{\omega}_I$, giving $\dot{R} = \widehat{\omega}_I R$.

For the relation between them, equate the two expressions: $R\widehat{\omega}_B = \widehat{\omega}_I R$, hence $\widehat{\omega}_I = R\widehat{\omega}_B R^\top$. Apply both sides to an arbitrary w and set $u = R^\top w$:

$$\widehat{\omega}_I w = R(\omega_B \times u) = (R\omega_B) \times (Ru) = (R\omega_B) \times w,$$

where the middle equality is the identity $R(a \times b) = (Ra) \times (Rb)$ valid for $R \in \text{SO}(3)$ (proved in Appendix A). Since w is arbitrary, $\omega_I = R\omega_B$. $\square$

Theorem 8.2 (Quaternion kinematics). *With $q(t)$ the unit quaternion of $R(t)$ and $\omega = \omega_B$ the body-frame angular velocity,*

$$\boxed{\dot{q} = \frac{1}{2} q \otimes (0, \omega)} \quad \text{equivalently} \quad \dot{q} = \frac{1}{2} \Omega(\omega) q, \quad \Omega(\omega) = \begin{pmatrix} 0 & -\omega_1 & -\omega_2 & -\omega_3 \\ \omega_1 & 0 & \omega_3 & -\omega_2 \\ \omega_2 & -\omega_3 & 0 & \omega_1 \\ \omega_3 & \omega_2 & -\omega_1 & 0 \end{pmatrix}.$$

If instead ω_I is used, the increment multiplies from the left: $\dot{q} = \frac{1}{2}(0, \omega_I) \otimes q$.

Proof. Over $[t, t+h]$ the attitude advances by a small body-frame rotation, so $q(t+h) = q(t) \otimes \delta q(h)$ where $\delta q(h)$ is the quaternion of the rotation by angle $|\omega|h$ about $\omega/|\omega|$. By Corollary 3.5,

$$\delta q(h) = \left(\cos \frac{|\omega|h}{2}, \sin \frac{|\omega|h}{2} \frac{\omega}{|\omega|} \right) = \left(1 - \frac{|\omega|^2 h^2}{8} + O(h^4), \frac{h}{2} \omega + O(h^3) \right),$$

using $\cos x = 1 - x^2/2 + O(x^4)$ and $\sin x = x + O(x^3)$ with $x = |\omega|h/2$. Therefore

$$\dot{q}(t) = \lim_{h \rightarrow 0} \frac{q(t) \otimes \delta q(h) - q(t)}{h} = q(t) \otimes \lim_{h \rightarrow 0} \frac{\delta q(h) - (1, 0)}{h} = q(t) \otimes (0, \frac{1}{2}\omega) = \frac{1}{2} q \otimes (0, \omega),$$

where pulling $q(t)$ out of the limit uses bilinearity and continuity of $\otimes$.

For the matrix form, expand $q \otimes (0, \omega)$ with Lemma 2.15:

$$q \otimes (0, \omega) = (-q \cdot \omega, q_0 \omega + q \times \omega).$$

Its scalar component is $-(q_1 \omega_1 + q_2 \omega_2 + q_3 \omega_3)$, which is precisely row 0 of $\Omega(\omega)q$. Its first vector component is $q_0 \omega_1 + (q_2 \omega_3 - q_3 \omega_2)$, and row 1 of $\Omega(\omega)q$ reads $\omega_1 q_0 + 0 \cdot q_1 + \omega_3 q_2 - \omega_2 q_3$ — the same. The second component is $q_0 \omega_2 + (q_3 \omega_1 - q_1 \omega_3)$ against row 2, $\omega_2 q_0 - \omega_3 q_1 + 0 \cdot q_2 + \omega_1 q_3$ — the same. The third is $q_0 \omega_3 + (q_1 \omega_2 - q_2 \omega_1)$ against row 3, $\omega_3 q_0 + \omega_2 q_1 - \omega_1 q_2 + 0 \cdot q_3$ — the same. Hence $q \otimes (0, \omega) = \Omega(\omega)q$ identically. $\square$

Proposition 8.3 (Norm is conserved by the exact flow). *If $\dot{q} = \frac{1}{2}\Omega(\omega)q$ then $\frac{d}{dt}|q|^2 = 0$.*

Proof. $\Omega(\omega)$ is skew-symmetric: inspect the displayed matrix — the diagonal is zero and each (μ, ν) entry is minus the (ν, μ) entry, e.g. $\Omega_{01} = -\omega_1$ while $\Omega_{10} = +\omega_1$, and $\Omega_{12} = \omega_3$ while $\Omega_{21} = -\omega_3$. Therefore

$$\frac{d}{dt}|q|^2 = \frac{d}{dt}(q^\top q) = 2q^\top \dot{q} = q^\top \Omega(\omega)q = 0,$$

because for any skew A and any vector x , $x^\top A x = (x^\top A x)^\top = x^\top A^\top x = -x^\top A x$, forcing $x^\top A x = 0$. $\square$

Pitfall: your integrator does not know about Proposition 8.3

The exact flow preserves $|q| = 1$, but a Runge–Kutta step does not: RK4 applied to $\dot{q} = \frac{1}{2}\Omega q$ leaves $|q| = 1 + O(h^5)$ per step, and the error accumulates monotonically because the truncated exponential of a skew matrix has norm slightly greater than one. Three standard responses, in increasing order of virtue:

1. **Renormalize** $q \leftarrow q/|q|$ after each step. Cheap, universal, and it silently converts a norm error into an attitude error — acceptable for most work, but note that you are projecting, not correcting.
2. **Add a Baumgarte term** $\dot{q} += \frac{1}{2}\lambda(1 - |q|^2)q$, which makes the unit sphere attracting rather than merely invariant. Tune λ against your step size.
3. **Integrate on the group.** Advance by the exact exponential of the increment, $q_{k+1} = q_k \otimes \exp(\frac{h}{2}(0, \bar{\omega}))$ with $\bar{\omega}$ a suitable midpoint average, so every iterate is a unit quaternion *by construction*. This is the Lie-group (Munthe–Kaas / Crouch–Grossman) philosophy: respect the manifold instead of correcting for having left it.

Option 3 is what a high-fidelity spacecraft simulator should do; option 1 is what most code does. Knowing that the difference exists, and being able to say why, is a good answer to “how would you improve our propagator?”

9 Estimation on $\text{SO}(3)$: why attitude filters are multiplicative

Proposition 9.1 (The constraint makes an additive covariance singular). *Let q be a random 4-vector supported on $\mathbb{S}^3$, with mean $\bar{q}$ and covariance $P = \mathbb{E}[(q - \bar{q})(q - \bar{q})^\top]$. Then to first order in the deviation, P is singular with $\bar{q}$ in its null space.*

Proof. Write $q = \bar{q} + \delta$ with $|\delta|$ small. The constraint gives

$$1 = |q|^2 = (\bar{q} + \delta)^\top (\bar{q} + \delta) = |\bar{q}|^2 + 2\bar{q}^\top \delta + |\delta|^2 = 1 + 2\bar{q}^\top \delta + O(|\delta|^2),$$

so $\bar{q}^\top \delta = O(|\delta|^2)$; to first order the deviation is orthogonal to the mean. Then

$$P\bar{q} = \mathbb{E}[\delta \delta^\top] \bar{q} = \mathbb{E}[\delta (\bar{q}^\top \delta)] = O(|\delta|^3) \approx 0,$$

so $\bar{q}$ is (to this order) a null vector of P and $\det P \approx 0$. □

Engineering translation: the multiplicative extended Kalman filter, and why it exists

Proposition 9.1 is fatal for a naive Kalman filter: the covariance you must invert in the gain computation is structurally singular, so the filter is numerically ill posed and will eventually produce a non-positive-definite P . The standard repair — the **MEKF** — is pure Lie theory in disguise:

1. Keep the attitude as a unit quaternion reference $\hat{q}$ that is *not* a filter state.
2. Represent the error *multiplicatively*, $q = \hat{q} \otimes \delta q$, and parameterize the small error by a three-vector, $\delta q \approx (1, \frac{1}{2}\delta\theta)$ — exactly the first-order expansion of Corollary 3.5 with $\theta = |\delta\theta|$ small.

3. Carry the 3×3 covariance of $\delta\theta$, which is non-singular because $\delta\theta$ ranges over the whole Lie algebra $\mathfrak{so}(3) \cong \mathbb{R}^3$ with no constraint.
4. After each update, **reset**: fold the estimated $\delta\theta$ into the reference, $\hat{q} \leftarrow \hat{q} \otimes \delta q$, renormalize, and zero the error state.

Stated in one sentence a mathematician will recognize instantly: *the state lives on the group, the uncertainty lives on the Lie algebra, and the exponential map moves between them.* That sentence is also the correct answer to “why can’t I just put a 4×4 covariance on the quaternion?” — and if you are asked to design an attitude estimator on a whiteboard, drawing that separation first is worth more than writing any filter equation.

Remark 9.2 (Uncertainty on the group, properly). The MEKF is the first-order version of a more honest object: a concentrated Gaussian on $\text{SO}(3)$, $q = \hat{q} \otimes \exp(\xi)$ with $\xi \sim \mathcal{N}(\mathbf{0}, \Sigma)$ on $\mathfrak{so}(3)$. Composition of uncertainties is then governed not by addition but by the Baker–Campbell–Hausdorff series,

$$\log(e^X e^Y) = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] - \frac{1}{12}[Y, [X, Y]] + \dots,$$

whose first correction $\frac{1}{2}[X, Y]$ is exactly the cross-product term $\frac{1}{2}\xi_1 \times \xi_2$ under Theorem 2.12. When two attitude uncertainties are compounded and the answer “looks rotated,” this bracket is why.

10 Interpolation: SLERP, derived

Proposition 10.1. *Let q_1, q_2 be unit quaternions with $\cos \Omega_{12} = \langle q_1, q_2 \rangle > 0$. The constant-speed geodesic on $\mathbb{S}^3$ from q_1 to q_2 is*

$$\text{slerp}(q_1, q_2; t) = \frac{\sin((1-t)\Omega_{12})}{\sin \Omega_{12}} q_1 + \frac{\sin(t\Omega_{12})}{\sin \Omega_{12}} q_2, \quad t \in [0, 1].$$

Proof. A great-circle arc through q_1 in the plane spanned by q_1, q_2 has the form $c(t) = \cos(t\Omega_{12})q_1 + \sin(t\Omega_{12})u$, where u is the unit vector in that plane orthogonal to q_1 . Gram–Schmidt gives

$$u = \frac{q_2 - \langle q_1, q_2 \rangle q_1}{|q_2 - \langle q_1, q_2 \rangle q_1|} = \frac{q_2 - \cos \Omega_{12} q_1}{\sin \Omega_{12}},$$

where the denominator is computed as $|q_2 - \cos \Omega_{12} q_1|^2 = 1 - 2 \cos^2 \Omega_{12} + \cos^2 \Omega_{12} = 1 - \cos^2 \Omega_{12} = \sin^2 \Omega_{12}$. Substituting,

$$c(t) = \cos(t\Omega_{12})q_1 + \sin(t\Omega_{12}) \frac{q_2 - \cos \Omega_{12} q_1}{\sin \Omega_{12}} = \frac{\cos(t\Omega_{12}) \sin \Omega_{12} - \sin(t\Omega_{12}) \cos \Omega_{12}}{\sin \Omega_{12}} q_1 + \frac{\sin(t\Omega_{12})}{\sin \Omega_{12}} q_2.$$

The numerator of the first coefficient is $\sin(\Omega_{12} - t\Omega_{12}) = \sin((1-t)\Omega_{12})$ by the sine subtraction formula. Finally $c(0) = q_1$ and $c(1) = q_2$, and $|c'(t)| = \Omega_{12}$ is constant, so the parameterization has constant speed. $\square$

Engineering translation: three rules for interpolating attitude

(i) If $\langle q_1, q_2 \rangle < 0$, replace q_2 by $-q_2$ first — same rotation (Theorem 4.4), shorter path. Skipping this makes a gimbal appear to take the scenic route through $> 180^\circ$. (ii) As $\Omega_{12} \rightarrow 0$ the formula divides by $\sin \Omega_{12} \rightarrow 0$; fall back to normalized linear interpolation, which agrees to $O(\Omega_{12}^2)$. (iii) SLERP is C^1 but not C^2 at the knots when chained, so a sequence of SLERP segments has discontinuous angular acceleration — fine for visualization, wrong for anything that then differentiates the signal to get a torque command.

11 The compendium: engineer’s questions, physicist’s answers

“Which convention are we using?” The four rows of the table in Section 1. Insist that the answer be written down in the interface control document, not inferred. State it as: scalar-first Hamilton, active, body-to-inertial.

“Why a half angle? It looks like a typo.” Because $SU(2) \rightarrow SO(3)$ is two-to-one (Theorem 4.4). The quaternion parameter must run at half speed so that a full 2π physical rotation corresponds to a 4π circuit of $\mathbb{S}^3$. It is not a normalization choice; it is forced by the topology.

“Why did the sign flip?” $\ker \varphi = \{\pm I\}$: q and $-q$ are the same rotation. Compare with $1 - |\langle q_1, q_2 \rangle|$, never $\|q_1 - q_2\|$.

“Why not Euler angles?” Two separate reasons, and it is worth keeping them distinct. *Practically*, gimbal lock: at $\pm 90^\circ$ pitch the first and third axes coincide, the Jacobian from Euler rates to ω becomes singular, and the rate command blows up. *Topologically*, $SO(3) \cong \mathbb{RP}^3$ is a compact 3-manifold, and no compact 3-manifold admits a global diffeomorphism onto an open subset of $\mathbb{R}^3$; therefore *no* three-parameter attitude representation can be globally non-singular. Euler angles are not badly chosen — three parameters are simply impossible. This is why every good representation is either four numbers with one constraint (quaternions) or nine with six constraints ($SO(3)$ matrices).

“ ω in body or inertial frame?” $\dot{R} = R\widehat{\omega}_B$ for body, $\dot{R} = \widehat{\omega}_I R$ for inertial, with $\omega_I = R\omega_B$ (Proposition 8.1). Right multiplication is the body frame; left multiplication is the inertial frame. Gyros measure body rates, so flight software almost always wants the right-multiply form. Getting this backwards produces an error that vanishes at $t = 0$ and grows — the worst kind, because the unit test passes.

“How do I test the implementation?” Test against mathematics, not against the previous implementation. The invariant list:

1. $\|R^\top R - I\|_\infty < 10^{-14}$ and $\det R = 1$ for every generated R .
2. $|q| = 1$ to machine precision; $q \otimes q^* = (1, \mathbf{0})$.
3. Round trip: $q \rightarrow R \rightarrow q'$ gives $q' = \pm q$ — assert on $|\langle q, q' \rangle| = 1$, and if your test asserts $q' = q$ it will fail intermittently and correctly.
4. Composition: $R(p \otimes q) = R(p)R(q)$. This is the single test that catches a convention mismatch.

5. Group action: $R(q)\mathbf{v}$ equals the vector part of $q \otimes (0, \mathbf{v}) \otimes q^*$.
 6. Known values: 90° about $\mathbf{e}_3$ sends $\mathbf{e}_1 \mapsto \mathbf{e}_2$.
 7. Small angle: $\|R(\theta\mathbf{n}) - (I + \theta\hat{\mathbf{n}})\| = O(\theta^2)$; verify the exponent numerically by halving θ and checking the error drops by four.
 8. Adversarial angles: $\theta = \pi$ exactly, $\theta = \pi \pm 10^{-8}$, $\theta = 0$, $\theta = 10^{-12}$. This is where Shepperd and the arccos/atan2 choice earn their keep.
- Items 3, 4 and 8 are the ones usually missing, and they are the ones that fail in flight.

“The propagator loses orthogonality overnight.” It always will; see the pitfall after Proposition 8.3. Either renormalize every step or integrate on the group.

“Can I average attitudes?” Not by averaging components — the mean of unit quaternions is not a unit quaternion, and the answer depends on the arbitrary signs. The correct object is the Fréchet mean: minimize $\sum_i d(q, q_i)^2$ with d the geodesic distance, computed either by the eigenvector of $\sum_i q_i q_i^\top$ belonging to the largest eigenvalue (Markley’s method) or by iterating $\hat{q} \leftarrow \hat{q} \otimes \exp(\frac{1}{N} \sum_i \log(\hat{q}^* \otimes q_i))$.

12 Reference card

Object	Formula
Hat map	$\hat{\mathbf{v}}\mathbf{w} = \mathbf{v} \times \mathbf{w}$, $\hat{\mathbf{v}}^2 = \mathbf{v}\mathbf{v}^\top - \mathbf{v} ^2 I$
Brackets	$[L_j, L_k] = \varepsilon_{jkl} L_l = [E_j, E_k]$, $E_j = -\frac{i}{2}\sigma_j$
Rodrigues	$e^{\theta\hat{\mathbf{n}}} = I + \sin\theta\hat{\mathbf{n}} + (1 - \cos\theta)\hat{\mathbf{n}}^2$
Axis-angle $\rightarrow q$	$q = (\cos\frac{\theta}{2}, \sin\frac{\theta}{2}\mathbf{n})$
Product	$p \otimes q = (p_0 q_0 - \mathbf{p} \cdot \mathbf{q}, p_0 \mathbf{q} + q_0 \mathbf{p} + \mathbf{p} \times \mathbf{q})$
Action	$\mathbf{v}' = (q_0^2 - \mathbf{q} ^2)\mathbf{v} + 2(\mathbf{q} \cdot \mathbf{v})\mathbf{q} + 2q_0(\mathbf{q} \times \mathbf{v})$
$q \rightarrow R$	$R = (q_0^2 - \mathbf{q} ^2)I + 2\mathbf{q}\mathbf{q}^\top + 2q_0\hat{\mathbf{q}}$
$R \rightarrow q$	Shepperd: t_μ largest of $\{1 + \text{tr } R, 1 \pm R_{11} \mp R_{22} \mp R_{33}, \dots\}$
Trace	$\text{tr } R = 1 + 2\cos\theta = 4q_0^2 - 1$
Angle	$\theta = 2 \text{atan2}(\mathbf{q} , q_0)$ (not arccos)
Kinematics	$\dot{q} = \frac{1}{2}q \otimes (0, \boldsymbol{\omega}_B)$, $\dot{R} = R\hat{\boldsymbol{\omega}}_B$
Error state	$q = \hat{q} \otimes \delta q$, $\delta q \approx (1, \frac{1}{2}\delta\boldsymbol{\theta})$, covariance on $\delta\boldsymbol{\theta} \in \mathbb{R}^3$
Distance	$d(q_1, q_2) = 2 \arccos \langle q_1, q_2 \rangle $

A The ε - δ identities used above

Lemma A.1. $\varepsilon_{ijk}\varepsilon_{ilm} = \delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}$.

Proof. Both sides are antisymmetric under $j \leftrightarrow k$ and under $l \leftrightarrow m$, and both vanish if $j = k$ or $l = m$. So it suffices to check one representative non-vanishing case and let the symmetries generate the rest. Take $j = l = 1$, $k = m = 2$. The left side is $\varepsilon_{i12}\varepsilon_{i12} = \varepsilon_{312}^2 = 1$ (only $i = 3$ contributes). The right side is $\delta_{11}\delta_{22} - \delta_{12}\delta_{21} = 1 \cdot 1 - 0 \cdot 0 = 1$. Now take $j = 1, k = 2, l = 2, m = 1$: the left is $\varepsilon_{i12}\varepsilon_{i21} = \varepsilon_{312}\varepsilon_{321} = (1)(-1) = -1$, and the right is $\delta_{12}\delta_{21} - \delta_{11}\delta_{22} = -1$. Finally if $\{j, k\} \neq \{l, m\}$ as sets, no value of i makes both ε factors non-zero, and correspondingly each δ product on the right contains a δ with distinct indices. All cases agree. $\square$

Corollary A.2 (Triple product). $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$.

Proof. Compute the j -th component, using $(\mathbf{x} \times \mathbf{y})_j = \varepsilon_{jkl}x_ky_l$ twice:

$$(\mathbf{a} \times (\mathbf{b} \times \mathbf{c}))_j = \varepsilon_{jkl}a_k(\mathbf{b} \times \mathbf{c})_l = \varepsilon_{jkl}a_k\varepsilon_{lmn}b_m c_n = \varepsilon_{ljk}\varepsilon_{lmn}a_k b_m c_n,$$

where the last step cycles $\varepsilon_{jkl} = \varepsilon_{ljk}$. Applying Lemma A.1 with the summed index l in first position,

$$= (\delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km})a_k b_m c_n = b_j(a_k c_k) - c_j(a_k b_k) = b_j(\mathbf{a} \cdot \mathbf{c}) - c_j(\mathbf{a} \cdot \mathbf{b}). \quad \square$$

Corollary A.3 (Rotations commute with the cross product). For $R \in \text{SO}(3)$ and all $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$, $R(\mathbf{a} \times \mathbf{b}) = (R\mathbf{a}) \times (R\mathbf{b})$.

Proof. Write the i -th component of the right side and insert $\det R = 1$ in the form of the classical identity $\varepsilon_{jkl}R_{jp}R_{kq}R_{lr} = \det R \varepsilon_{pqr} = \varepsilon_{pqr}$:

$$((R\mathbf{a}) \times (R\mathbf{b}))_i = \varepsilon_{ijk}R_{jp}a_p R_{kq}b_q.$$

Multiply and divide by the orthogonality relation $R_{li}R_{lm} = \delta_{im}$:

$$= R_{li} \left(R_{li}\varepsilon_{ijk}R_{jp}R_{kq} \right) a_p b_q \Big|_{\text{sum over } i} \quad \text{where} \quad \varepsilon_{ijk}R_{li}R_{jp}R_{kq} = \varepsilon_{lpq}.$$

Hence $((R\mathbf{a}) \times (R\mathbf{b}))_i = R_{li}^{-1\top} \varepsilon_{lpq} a_p b_q$; since $R^{-\top} = R$ for $R \in \text{SO}(3)$, this is $R_{il} \varepsilon_{lpq} a_p b_q = R_{il}(\mathbf{a} \times \mathbf{b})_l = (R(\mathbf{a} \times \mathbf{b}))_i$. $\square$

Remark A.4. Corollary A.3 fails for $R \in \text{O}(3)$ with $\det R = -1$: the identity picks up the factor $\det R$, so a reflection sends $\mathbf{a} \times \mathbf{b}$ to $-(R\mathbf{a}) \times (R\mathbf{b})$. This is the precise sense in which the cross product is a pseudovector, and the practical reason a left-handed coordinate convention silently negates every angular quantity in a simulation.